LESSON PLAN · SCRATCH (GR 3–5) BEGINNER: SCRATCH GAME MAKERS

Lesson 5: Forever Loops and Costume Animation

Module 2 | 45 minutes | Loops (forever) and animation | Grades 3–5 · ages 8–10

Objective: Use forever loops, costumes and wait blocks to animate a sprite, and make it walk and bounce off the edges.

Success criteria

- I can make a sprite animate with costumes and a forever loop.
- I can explain why a wait is needed.
- I can choose between repeat and forever.

Key vocabulary

- **forever loop**: A loop that never stops by itself.
- **costume**: One picture of a sprite.
- **animation**: Pictures changing quickly to look like movement.
- **bounce**: Turning around at the edge of the stage.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Sticky-note flip book.
5 min	**Watch**	Lesson video; pause at “Why wait?”.
8 min	**Build along**	Walking sprite with costumes, wait and bounce.
15 min	**Perform**	Forever-forward puzzle, walking animation challenge, repeat-or-forever sort, predict.
8 min	**Probe**	Fix the flapping bird; repeat vs forever writing.
4 min	**Wrap-up**	Exit ticket; animation gallery on the projector.

Unplugged warm-up

Flip-book warm-up (5 min): draw a stick figure in two poses on two sticky notes. Flip between them quickly. Then show what happens if you flip VERY fast (blur) or VERY slow (no animation).

Teaching the idea

A forever loop repeats until the program is stopped; games and animations rely on it. Animation = switching costumes with a short pause. “if on edge, bounce” reverses direction at the stage edge; rotation style left-right prevents upside-down sprites.

Analogy: Animation is a flip book; a forever loop is someone flipping the pages over and over.

Live demo: Show forever{next costume} with no wait — it flickers. Add wait (0.2). Then add move + bounce; let it walk upside down and ask students how to fix it.

Questions to ask

Ask	Listen for
What happens without a wait?	It changes too fast to see.
When would you use forever instead of repeat?	When it should keep going for the whole game.
How do you stop a forever loop?	The red stop sign (or a stop block).

Differentiation

Support
Provide the walking script on a card; students change only the numbers and the sprite.

Stretch
Make two sprites animate at different speeds, or make a sprite grow and shrink forever using change size.

Common misconceptions

Misconception	How to fix it
Forever loops are bad because they never stop	They are useful for things that should keep going; the stop sign ends them.
Next costume needs a costume name	“next costume” just goes to the next picture; it loops back to the first.
Blocks after a forever loop will run	Nothing after a forever loop ever runs — that is why it has no notch at the bottom.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

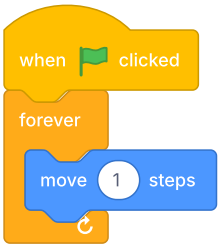
Exit ticket
Draw the three blocks you need inside a forever loop to make a sprite walk.

Homework
Make a two-page flip book at home and explain to someone how it is like a forever loop.

Answer key — Kitabash activities

1. Forever forward (perform · block code)

One solution:



2. Code Lab: walking animation (perform · code lab)

Goals checked: Uses a forever loop · Changes costume · Has a wait so we can see it · Walks across the stage · Bounces at the edge

One solution:

Runner



3. Repeat or forever? (perform · categorize)

repeat (a set number of times): Walk across the stage exactly 3 times, Flap wings 10 times, Draw a hexagon
forever (until you stop it): Background music for the whole game, A clock hand that never stops, Keep checking for the space key during the game

4. Predict it (perform · multiple choice)

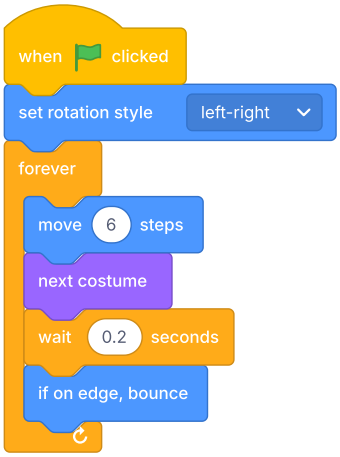
Answer: Walk back and forth across the stage, never stopping

5. Code Lab: fix the flapping bird (probe · code lab)

Goals checked: Flaps at a speed we can see · Flies across the stage · Keeps flapping · Turns around at the edges

One solution:

Bird



6. Explain your thinking (probe · free response)

Look for: repeat, forever, stop. **Example:** Repeat (10) runs the blocks 10 times and then stops, like flapping wings 10 times. Forever never stops until you press stop, like background music in a game.

Answer key — worksheet

1. **Flip book:** Two drawings with a small change between them.
2. **Repeat or forever?:** F · R · F · R
3. **Fix the bug:** wait (0.2) seconds — so we can see it; move (5) steps (and if on edge, bounce) so it walks.
4. **Order the walk:** Any order that includes all four works; a common one is move, next costume, wait, bounce.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.